PS C:\Users\tg070\OneDrive\Área de Trabalho\ST SF\Univesp\estoque_python> py -m pip install fastapi uvicorn jinja2

```
Collecting fastapi
Downloading fastapi-0.135.1-py3-none-any.whl.metadata (30 kB)
Collecting uvicorn
Downloading uvicorn-0.41.0-py3-none-any.whl.metadata (6.7 kB)
Collecting jinja2
Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting starlette>=0.46.0 (from fastapi)
Downloading starlette-0.52.1-py3-none-any.whl.metadata (6.3 kB)
Collecting pydantic>=2.7.0 (from fastapi)
Downloading pydantic-2.12.5-py3-none-any.whl.metadata (90 kB)
Collecting typing-extensions>=4.8.0 (from fastapi)
Downloading typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Collecting typing-inspection>=0.4.2 (from fastapi)
Downloading typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting annotated-doc>=0.0.2 (from fastapi)
Downloading annotated_doc-0.0.4-py3-none-any.whl.metadata (6.6 kB)
Collecting click>=7.0 (from uvicorn)
Downloading click-8.3.1-py3-none-any.whl.metadata (2.6 kB)
Collecting h11>=0.8 (from uvicorn)
Downloading h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Collecting MarkupSafe>=2.0 (from jinja2)
Downloading markupsafe-3.0.3-cp314-cp314-win_amd64.whl.metadata (2.8 kB)
Collecting colorama (from click>=7.0->uvicorn)
Downloading colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Collecting annotated-types>=0.6.0 (from pydantic>=2.7.0->fastapi)
Downloading annotated_types-0.7.0-py3-none-any.whl.metadata (15 kB)
Collecting pydantic-core==2.41.5 (from pydantic>=2.7.0->fastapi)
Downloading pydantic_core-2.41.5-cp314-cp314-win_amd64.whl.metadata (7.4 kB)
Collecting anyio<5,>=3.6.2 (from starlette>=0.46.0->fastapi)
Downloading anyio-4.12.1-py3-none-any.whl.metadata (4.3 kB)
Collecting idna>=2.8 (from anyio<5,>=3.6.2->starlette>=0.46.0->fastapi)
Downloading idna-3.11-py3-none-any.whl.metadata (8.4 kB)
Downloading fastapi-0.135.1-py3-none-any.whl (116 kB)
Downloading uvicorn-0.41.0-py3-none-any.whl (68 kB)
Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
```

```
Downloading annotated_doc-0.0.4-py3-none-any.whl (5.3 kB)
Downloading click-8.3.1-py3-none-any.whl (108 kB)
Downloading h11-0.16.0-py3-none-any.whl (37 kB)
Downloading markupsafe-3.0.3-cp314-cp314-win_amd64.whl (15 kB)
Downloading pydantic-2.12.5-py3-none-any.whl (463 kB)
Downloading pydantic_core-2.41.5-cp314-cp314-win_amd64.whl (2.0 MB)
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.0/2.0 MB 5.3 MB/s 0:00:00
Downloading annotated_types-0.7.0-py3-none-any.whl (13 kB)
Downloading starlette-0.52.1-py3-none-any.whl (74 kB)
Downloading anyio-4.12.1-py3-none-any.whl (113 kB)
Downloading idna-3.11-py3-none-any.whl (71 kB)
Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Downloading typing_inspection-0.4.2-py3-none-any.whl (14 kB)
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Installing collected packages: typing-extensions, MarkupSafe, idna, h11, colorama, annotated-
types, annotated-doc, typing- ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ ━━ 15/16 [fastapi]
WARNING: The script fastapi.exe is installed in
'C:\Users\tg070\AppData\Local\Python\pythoncore-3.14-64\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-
script-location.
Successfully installed MarkupSafe-3.0.3 annotated-doc-0.0.4 annotated-types-0.7.0 anyio-4.12.1
click-8.3.1 colorama-0.4.6 fastapi-0.135.1 h11-0.16.0 idna-3.11 jinja2-3.1.6 pydantic-2.12.5 pydantic-
core-2.41.5 starlette-0.52.1 typing-extensions-4.15.0 typing-inspection-0.4.2 uvicorn-0.41.0

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: C:\Users\tg070\AppData\Local\Python\pythoncore-3.14-64\python.exe
-m pip install --upgrade pip
```

✅ **PERFEITO! INSTALAÇÃO CONCLUÍDA 100%!** 🎉

FastAPI + Uvicorn + Jinja2 instalados com sucesso!

🚀 PRÓXIMOS 3 PASSOS (no VS Code Terminal):

1. CRIE o BANCO + PRODUTOS:

```
# No terminal VS Code (Ctrl+`)
python init_db.py
```

Resultado esperado:

```
✅ Banco criado! 5 produtos inseridos!
```

2. COLE os 5 ARQUIVOS (main.py, init_db.py, templates/):

```
✅ main.py                ← Backend + Frontend
✅ init_db.py            ← Banco SQLite
✅ templates/dashboard.html
✅ templates/venda_form.html
✅ static/style.css (opcional)
```

3. RODE o SISTEMA:

```
uvicorn main:app --reload
```

Resultado esperado:

```
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
```

✅ ABRA NAVEGADOR:

```
http://localhost:8000
```

VOCÊ VERÁ:

```
□ Dashboard Estoque MEI
✅ 5 produtos com estoque
✅ Cards KPI (Total Vendas, Baixo Estoque)
✅ Botão "□ Nova Venda"
✅ API teste funcionando
```

□ TESTE COMPLETO:

```
1. ✅ Dashboard: http://localhost:8000
2. ✅ Nova Venda: Clique botão → Selecione Arroz → Qtd 2 → Confirmar
3. ✅ Volta Dashboard → Estoque Arroz = 48 (-2)
4. ✅ API: http://localhost:8000/api/produtos → JSON igual Laravel!
5. ✅ Docs: http://localhost:8000/docs → Swagger automático!
```

⚠ AVISO pip (ignora):

```
WARNING: Scripts not in PATH ← Normal no Windows
✅ Funciona perfeitamente mesmo assim!
```

⚡ COMANDOS EXATOS (copie agora):

```
# 1. Terminal VS Code (Ctrl+`)
python init_db.py
# ✅ "Banco criado!"

# 2. uvicorn main:app --reload
# ✅ "Uvicorn running on http://127.0.0.1:8000"

# 3. Navegador: http://localhost:8000
```

FastAPI 0.135.1 instalado = SISTEMA PYTHON PRONTO! ⚡

Seu Laravel agora é Python em 2 minutos! ⚡⚡

Univesp: "Migração Laravel→FastAPI = Tecnologia de ponta!" ⚡

Cole arquivos → `python init_db.py` → `uvicorn main:app` = Dashboard funcionando! ⚡

⁂

1. https://stackoverflow.com/questions/73562342/getting-exception-has-occurred-assertionerror-jinja2-must-be-installed-to-use
2. https://fastapi.tiangolo.com/advanced/templates/
3. https://talentmp.com/blog/fastapi-with-jinja2-templates-a-1761200081261
4. https://developer-service.blog/how-to-build-dynamic-frontends-with-fastapi-and-jinja2/
5. https://realpython.com/fastapi-jinja2-template/
6. https://www.geeksforgeeks.org/python/install-fastapi-and-run-your-first-fastapi-server-on-windows/
7. https://www.youtube.com/watch?v=uMe7BgmhuDg
8. https://fastapi.tiangolo.com/deployment/manually/
9. https://www.tabnews.com.br/ricardotenv/renderizando-html-dinamico-com-fastapi-e-jinja2-uma-alternativa-simples-aos-spas
10. https://code.visualstudio.com/docs/python/tutorial-fastapi